

Qatar-4b

R-2535

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211015 · created 2026-08-25T23:31:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459502.7117 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.023
Transit depth (Rp/Rs)^2	3.38 +/- 0.83 [%]
Orbital Inclination (inc)	83.63 +/- 0.93
Airmass coefficient 1 (a1)	1.162 +/- 0.03
Airmass coefficient 2 (a2)	-0.127 +/- 0.053
Scatter in the residuals of the lightcurve fit is	2.65 %
Best Comparison Star	#1 - [81, 68]
Optimal Method	PSF photometry
Transit Duration (day)	0.0708 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.023 vs 0.138 (deviation 1.56σ)
Duration fit vs published	0.0708 d vs 0.0892 d (deviation 1.44σ)
Mid-time O-C	-6.4 min
Depth SNR	6.2
Residual scatter	2.65 %

Light curve

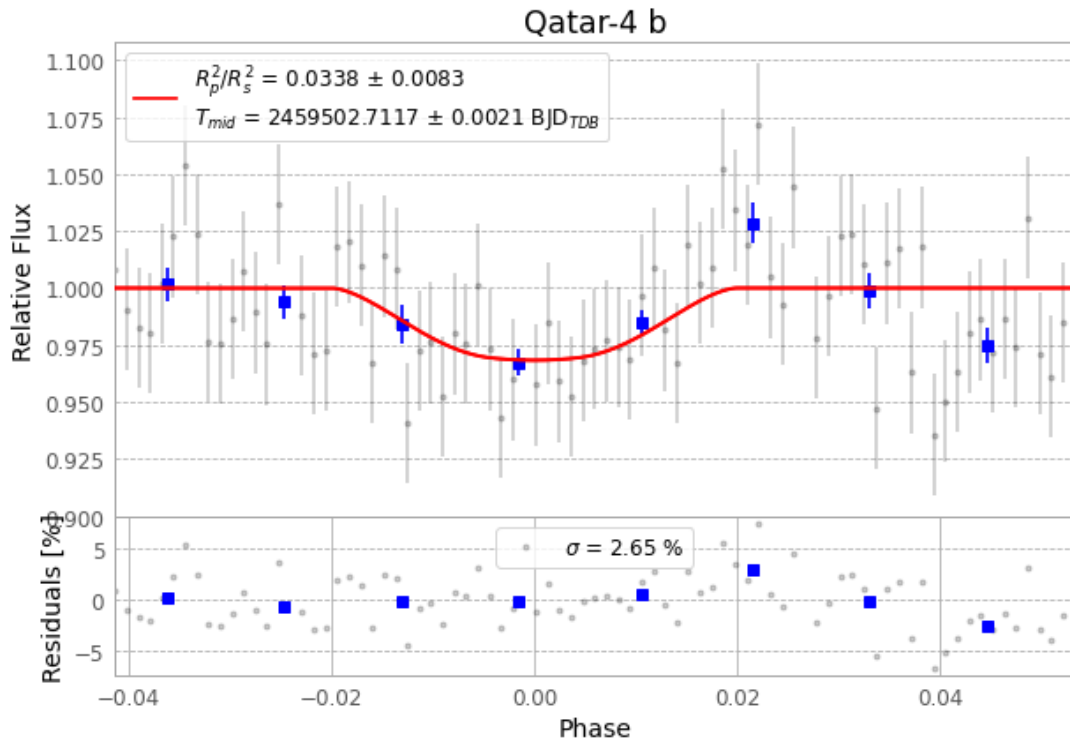


Figure 1 — FinalLightCurve_Qatar-4 b_2021-10-14.png (SHA-256 cb043d05b320cb5c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8000b35acd4fc97b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

83 FITS frames (55 MB), Qatar-4211015030912.FITS → Qatar-4211015071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211015.log (0bb388be81f2...), FinalLightCurve_Qatar-4 b_2021-10-14.png (cb043d05b320...), FinalLightCurve_Qatar-4 b_2021-10-14.csv (e930e48a0563...)

Compute resources

Wall time 123.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:31Z.